Unit 4: Virtualization and Cloud Computing (5 Hours)

Er. Sandesh Thapa Magar

Everest Engineering College

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Virtualization vs Physical Infrastructure

S.N.	Virtualization	Physical Infrastructure
1	It is a technology that creates multiple virtual machines (VMs) on a single physical system.	It refers to actual hardware resources like servers, storage, and networking devices.
2	Resources are shared and optimized using software (hypervisor).	Resources are directly used by physical systems without abstraction.
3	Multiple operating systems can run on a single physical machine.	Usually one operating system runs per server.
4	Highly scalable and flexible.	Limited scalability; requires new hardware for expansion.
5	Efficient utilization of CPU, memory, and storage.	Often leads to underutilization of resources.
6	Lower cost in long-term due to resource sharing.	Higher cost due to hardware, maintenance, and energy usage.
7	Provides isolation between virtual machines.	No virtualization-based isolation; systems are independent hardware units.
8	Example: VMware, VirtualBox, Hyper-V.	Example: Physical servers, data centers, networking hardware.

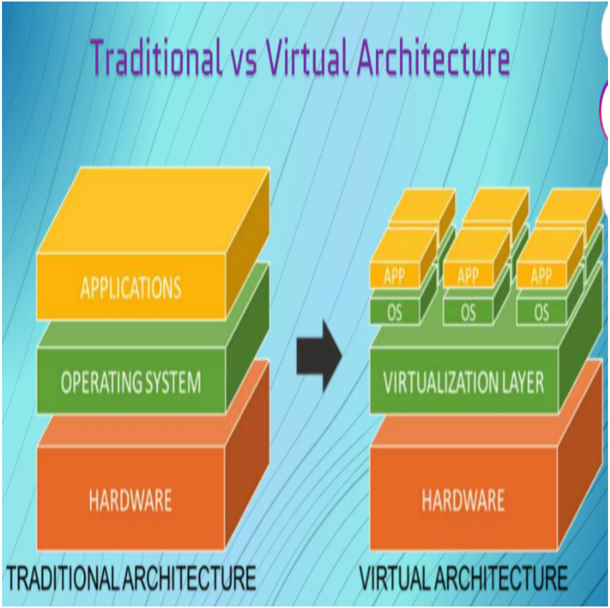
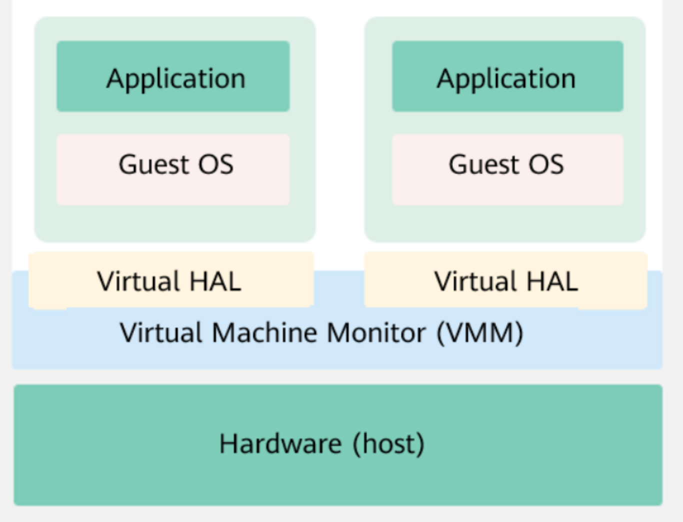


Figure: virtualization

Virtualization

- Virtualization is the fundamental technology that powers Cloud Computing.
- It allows multiple simulated environments called Virtual Machines (VMs) to be created from a single physical hardware system.
- Before virtualization, one physical server could run only one operating system and one main task.
- This led to poor resource utilization, for example, 90% CPU may remain idle if only 10% is used.
- Virtualization solves this by allowing one physical server to host multiple virtual servers.
- Each virtual server runs its own operating system and applications independently.
- All virtual machines are isolated from each other even though they share the same physical hardware.


Application
Guest OS
Application
Guest OS
Virtual HAL
Virtual HAL
Virtual Machine Monitor (VMM)
Hardware (host)

Core Architecture of Virtualization

- The core of virtualization is the **Hypervisor**.
- **Physical Hardware (Host):** The actual physical server that includes CPU, RAM, storage, and other hardware resources.
- **Hypervisor:** A lightweight software layer that sits between physical hardware and virtual machines. It manages and allocates resources to VMs (e.g., assigning 2GB RAM to a VM).
- **Virtual Machine (Guest):** A software-based computer that behaves like a real machine. It has its own operating system, libraries, and applications.
- Each VM operates independently but shares the same physical hardware through the hypervisor.

Working of Virtualization

- Virtualization uses special software called a **Hypervisor** to create multiple virtual computers on a single physical machine.
- Virtual machines (VMs) behave like real computers but share the same physical hardware.
- The physical computer is called the **Host**, while the virtual machines are called **Guests**.
- A single host can run multiple guest virtual machines at the same time.
- Each VM can have its own operating system, which may be the same or different from the host OS.
- Each virtual machine works like a standalone computer with its own settings, programs, and configuration.
- VMs access shared resources such as CPU, RAM, and storage but behave as if they have dedicated hardware.

Advantages of Virtualization (Part 1)

- **Efficient Resource Utilization**
  A single physical server can run multiple VMs. CPU, RAM, and storage are shared, reducing idle hardware usage and improving efficiency.
- **Cost Reduction**
  Fewer physical servers are required, reducing hardware cost, electricity usage, cooling systems, and maintenance expenses.
- **Easy Scalability**
  New virtual machines can be created quickly without purchasing new hardware, and removed when not needed.

Advantages of Virtualization (Part 2)

- **Isolation Between Systems**
  Each VM is independent. If one VM fails, others are not affected, improving stability and security.
- **Improved Backup and Recovery**
  VMs can be saved as images and restored quickly in case of failure, reducing downtime.
- **Support for Multiple Operating Systems**
  Different OS like Windows and Linux can run on the same physical machine simultaneously.

Advantages of Virtualization (Part 3)

- **Better Disaster Recovery**
  VMs can be moved or restored on another server quickly using live migration.
- **Energy Efficiency**
  Fewer physical servers reduce power consumption and cooling requirements, making it environmentally friendly.

Disadvantages of Virtualization (Part 1)

- **Performance Overhead**
  - Virtual machines do not run directly on hardware
  - Hypervisor adds extra processing layer
  - Slightly lower performance than physical systems
- **High Initial Setup Cost**
  - Requires powerful servers and storage systems
  - Virtualization software may require licenses
  - Initial investment is high

Disadvantages of Virtualization (Part 2)

- **Complex Management**
  - Managing multiple VMs requires skilled administrators
  - Monitoring and resource allocation is complex
- **Single Point of Failure**
  - If host machine fails, all VMs go down
  - Needs backup or clustering for protection
- **Security Risks**
  - Hypervisor is a critical layer
  - If compromised, all VMs may be affected

Disadvantages of Virtualization (Part 3)

- **Resource Contention**
  - Multiple VMs share CPU, RAM, and disk
  - Overloading causes slow performance
- **Software Licensing Issues**
  - Each VM may need separate license
  - Increases cost in enterprise systems
- **Dependence on Powerful Hardware**
  - Requires high-performance CPU and RAM
  - Low-end systems cannot run efficiently

Hypervisor (VMM)

A hypervisor (also called Virtual Machine Monitor – VMM) is software, firmware, or hardware that creates and manages Virtual Machines (VMs). It allows multiple operating systems to run on a single physical computer by sharing hardware resources like CPU, memory, and storage.

Example: One physical server can run:

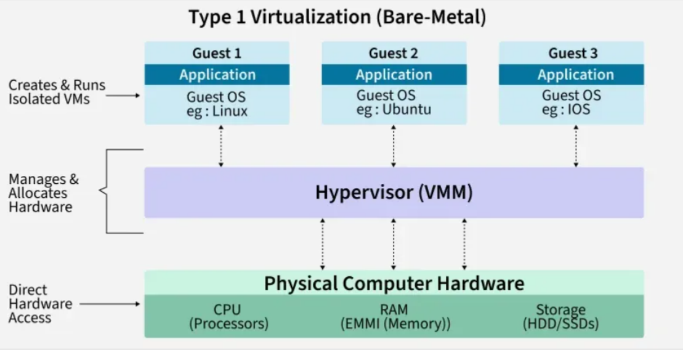
- Windows VM
- Linux VM
- macOS VM

all at the same time using a hypervisor.

Types of Hypervisors

There are two main types:

- Type 1 Hypervisor (Bare-Metal)
- Type 2 Hypervisor (Hosted)


Type 1 Virtualization (Bare-Metal)
Creates & Runs Isolated VMs
Guest 1
Application
Guest OS eg : Linux
Guest 2
Application
Guest OS eg : Ubuntu
Guest 3
Application
Guest OS eg : IOS
Manages & Allocates Hardware
Hypervisor (VMM)
Direct Hardware Access
Physical Computer Hardware
CPU (Processors)
RAM (EMMI (Memory))
Storage (HDD/SSDs)

Type 1 Hypervisor (Bare-Metal)

Runs directly on hardware without a host operating system.

Features:

- Faster performance
- More secure
- Used in data centers and cloud computing

Examples:

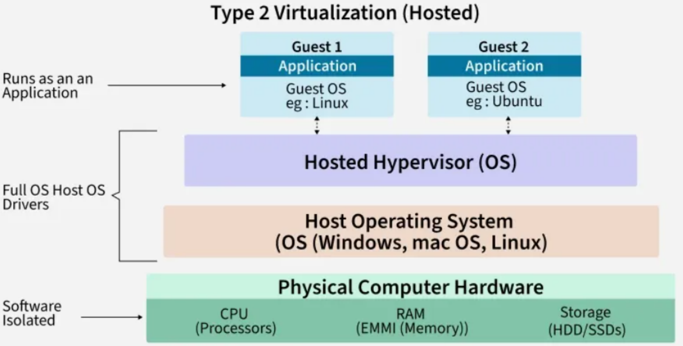
- VMware ESXi
- Microsoft Hyper-V

Advantages:

- High efficiency
- Better resource management
- Enterprise-grade security

Disadvantages:

- More complex setup
- Requires dedicated hardware


Type 2 Virtualization (Hosted)
Guest 1
Application
Guest OS
eg : Linux
Guest 2
Application
Guest OS
eg : Ubuntu
Runs as an an
Application
Hosted Hypervisor (OS)
Full OS Host OS
Drivers
Host Operating System
(OS (Windows, mac OS, Linux)
Physical Computer Hardware
Software
Isolated
CPU
(Processors)
RAM
(EMMI (Memory))
Storage
(HDD/SSDs)

Type 2 Hypervisor (Hosted)

Runs on top of an existing operating system.

Examples:

- Oracle VM VirtualBox
- VMware Workstation
- Parallels Desktop

Advantages:

- Easy to install and use
- Good for testing and learning
- Runs on personal computers

Disadvantages:

- Slower than Type 1
- Depends on host OS performance

Type System

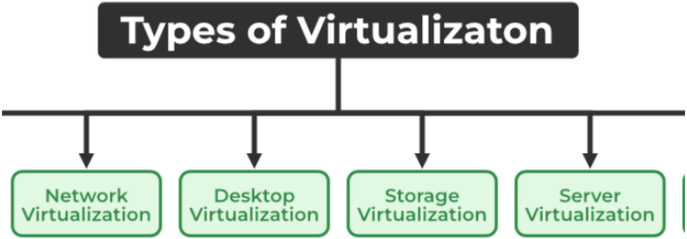


Figure: Hybrid Computer System Structure

Network Virtualization

Network Virtualization is the process of creating a virtual (software-based) version of a network that operates independently of the physical network hardware.

It allows multiple virtual networks to run on the same physical infrastructure.

Example of Network Virtualization

A university has one physical network infrastructure.
Using network virtualization, it can create:

- Virtual Network 1 for Students
- Virtual Network 2 for Faculty
- Virtual Network 3 for Administration

Although all use the same physical cables and switches, they operate separately and securely.

How Network Virtualization Works

- Physical network resources (switches, routers, links) are abstracted.
- Software creates virtual networks on top of the physical network.
- Each virtual network is assigned its own configuration and policies.
- Users and applications communicate through the virtual network as if it were a dedicated physical network.

Components of Network Virtualization

1. **Virtual Switch (vSwitch)**
   - Software-based switch
   - Connects virtual machines within a host
2. **Virtual Router**
   - Routes traffic between virtual networks
   - Functions like a physical router

Components of Network Virtualization (Continued)

3. Hypervisor

- Creates and manages virtual machines
- Manages virtual networking
- Examples: VMware ESXi, Microsoft Hyper-V

4. SDN Controller

- Centralized software for managing virtual networks
- Commonly used in Software-Defined Networking (SDN)

Network Virtualization Architecture

Applications / Users

↓

Virtual Networks

↓

Virtual Switches / Virtual Routers

↓

Hypervisor / SDN Controller

↓

Physical Network Infrastructure
(Switches, Routers, Cables)

Desktop Virtualization

Desktop Virtualization is a technology that separates a desktop environment (operating system, applications, and user data) from the physical computer.

The desktop runs on a remote server or virtual machine and is accessed through a network.

Example

A college computer lab has 100 computers.
Instead of installing software on all 100 computers individually:

- Desktops are created on a central server.
- Students log in from any computer.
- Their desktop, files, and applications appear instantly.

How It Works

- A server hosts multiple virtual desktops.
- Each user is assigned a virtual desktop.
- Users connect through a network.
- The server processes tasks and sends the display to the user's device.

Components of Desktop Virtualization

1. **Hypervisor**
   - Creates and manages virtual desktops.
   - Examples:
     - VMware ESXi
     - Microsoft Hyper-V
2. **Virtual Machines (VMs)**
   - Run individual desktop operating systems.

Components of Desktop Virtualization (Continued)

3. Connection Broker

- Authenticates users.
- Connects users to their virtual desktops.

4. Client Device

The device used to access the virtual desktop:

- PC
- Laptop
- Tablet
- Thin Client

Desktop Virtualization Architecture

Users

↓

Client Devices

(PC / Laptop / Tablet / Thin Client)

↓

Connection Broker

↓

Virtual Desktops (VMs)

↓

Hypervisor

↓

Physical Server

Storage Virtualization

Storage Virtualization is the process of combining multiple physical storage devices into a single virtual storage system. It hides the complexity of physical storage and presents it as one centralized storage resource to users and applications.

Example

A company has:

- Hard Disk 1 = 500 GB
- Hard Disk 2 = 1 TB
- Hard Disk 3 = 2 TB

Using storage virtualization, these disks can be combined and presented as a single virtual storage pool.

Applications and users see one storage system instead of multiple separate disks.

How It Works

- Physical storage devices are connected to the system.
- A virtualization layer is placed between storage hardware and applications.
- Physical storage resources are combined into a virtual pool.
- Virtual storage volumes are created from the pool.
- Applications access virtual volumes without knowing the actual physical location of data.

Server Virtualization

Server Virtualization is the process of dividing a single physical server into multiple virtual servers using virtualization software called a hypervisor. Each virtual server (called a Virtual Machine) works independently and can run its own operating system and applications.

Example

A company has one powerful physical server.
Using server virtualization, it can create:

- Virtual Server 1 → Linux Web Server
- Virtual Server 2 → Windows Database Server
- Virtual Server 3 → File Server

All of these run on the same physical machine but behave as separate independent servers.

How It Works

- A physical server (host machine) is installed with a hypervisor.
- The hypervisor divides hardware resources like CPU, RAM, and storage.
- Multiple Virtual Machines (VMs) are created.
- Each VM runs its own operating system.
- All VMs share the same physical hardware but operate independently.

Virtualization Platforms (Overview)

A virtualization platform is a software system that allows a single physical computer to run multiple Virtual Machines (VMs) at the same time using a hypervisor.
It creates a complete environment where multiple operating systems can run independently on the same hardware.

How a Virtualization Platform Works

A virtualization platform sits between the hardware and virtual machines.

Working Steps:

- Physical computer (host machine) starts
- Virtualization software is installed
- Hypervisor manages hardware resources
- Virtual Machines (VMs) are created
- Each VM is allocated:
  - CPU
  - RAM
  - Storage
  - Network
- Each VM runs its own operating system

Result: Each VM behaves like a separate computer.

Functions of Virtualization Platform

1. **VM Creation**
   - Creates multiple virtual computers on one physical system
2. **Resource Allocation**
   - Divides hardware resources:
     - CPU
     - RAM
     - Disk space
     - Network
3. **Isolation**
   - Each VM is independent
   - Failure of one VM does not affect others
4. **Management Tools**
   - Start / stop VMs
   - Install operating systems
   - Take snapshots
   - Clone or delete VMs

What is VMware?

VMware is a virtualization software platform that creates a virtual layer between hardware and operating systems. This layer is called a hypervisor, which enables multiple operating systems to run simultaneously on the same physical machine.

Each Virtual Machine has its own:

- Operating system
- CPU allocation
- Memory (RAM)
- Storage space

How VMware Works

- VMware is installed on a physical machine (host)
- It creates a hypervisor layer
- Hypervisor manages hardware resources
- Multiple Virtual Machines (VMs) are created
- Each VM runs independently

Result: Efficient hardware utilization and strong isolation between systems.

Popular VMware Products

VMware Workstation

- Runs on desktop/laptop computers
- Used for development and testing
- Supports multiple operating systems

VMware ESXi

- Bare-metal hypervisor (installed directly on hardware)
- Used in servers and data centers
- High performance and stability

VMware vSphere

- Complete virtualization suite
- Used for cloud and data center management
- Provides VM monitoring and load balancing

Features of VMware

1. **High Performance Virtualization**
   - Efficient resource allocation
   - Near-native VM performance
2. **Snapshots**
   - Saves VM state
   - Allows rollback to previous state
3. **Security & Isolation**
   - Each VM is isolated
   - One VM failure does not affect others
4. **Advanced Networking**
   - Virtual switches and routers
   - Custom network configuration
5. **Live Migration**
   - Move running VMs between servers without stopping

Oracle VirtualBox

Oracle VirtualBox is a free and open-source virtualization software that allows users to run multiple operating systems on a single physical computer. It is widely used for learning, testing, and software development.

What is VirtualBox?

- A hosted hypervisor (Type 2 hypervisor)
- Runs on Windows, Linux, macOS, and other OS
- Allows multiple guest operating systems on one host system

How VirtualBox Works

- Installed on a host operating system
- Creates a virtual layer on top of the host OS
- Shares hardware resources (CPU, RAM, storage)
- Each virtual machine runs independently

Example:

- Host OS: Windows 10
- Guest OS: Linux Ubuntu / Windows Server

Key Features of VirtualBox

1. **Free and Open Source**
   - No licensing cost
   - Suitable for students and developers
2. **Cross Platform Support**
   - Works on Windows, Linux, macOS, Solaris
3. **Snapshots**
   - Save VM state at any time
   - Restore previous state if needed
4. **Cloning**
   - Create duplicate virtual machines easily
5. **Networking Options**
   - NAT
   - Bridged networking
   - Host-only networking
6. **Shared Folders**
   - Easy file sharing between host and guest OS

Microsoft Hyper-V

Microsoft Hyper-V is a native virtualization platform developed by Microsoft that allows users to create and manage Virtual Machines (VMs) on Windows systems and servers. It is widely used in enterprise environments, cloud computing, and server virtualization.

What is Hyper-V?

- A Type 1 hypervisor (bare-metal hypervisor)
- Runs directly on hardware, not on top of an OS
- Enables multiple operating systems on a single machine

How Hyper-V Works

- Installed on Windows (Pro, Enterprise, or Server)
- Directly manages hardware resources (CPU, RAM, storage)
- Creates isolated Virtual Machines (VMs)
- Each VM runs its own operating system independently

Advantage: Higher performance and security due to Type-1 architecture.

Key Features of Hyper-V

1. **Native Hypervisor**
   - Runs directly on hardware
   - No separate host OS layer
2. **Virtual Machine Support**
   - Supports Windows, Linux, and other OS
   - Fully isolated virtual machines
3. **Live Migration**
   - Move running VMs between servers without downtime
4. **Virtual Networking**
   - Virtual switches for VM communication
   - Supports advanced network configurations
5. **Checkpoints (Snapshots)**
   - Save VM state
   - Restore previous state when needed
6. **Cloud Integration**
   - Works with Microsoft Azure for hybrid cloud systems

Introduction to Cloud Computing

Cloud computing has transformed how organizations build, deploy, and scale technology.

- Access computing resources over the Internet on demand.
- Eliminates the need for physical servers and upfront hardware costs.
- Provides computing power and storage whenever needed.
- Supports scalability, flexibility, and cost efficiency.

Problem Before Cloud Computing

The Old Way (On-Premises)

- Companies purchased physical servers, storage, and networking equipment.
- Required large upfront investment (Capital Expenditure - CapEx).
- Future demand had to be estimated in advance.
- Often led to overprovisioning and wasted resources.
- Setting up new infrastructure could take weeks or months.

Cloud Computing Solution

The New Way (Cloud Computing)

- Computing resources are rented from cloud providers.
- Pay-as-you-go model reduces upfront costs.
- Resources can be provisioned instantly.
- Scales up or down based on demand.
- Eliminates overprovisioning and resource wastage.

Characteristics of Cloud Computing

1. **On-Demand Self-Service**
    - Users access resources automatically when needed.
    - Example: Creating a virtual machine in minutes.
2. **Broad Network Access**
    - Services accessible via the Internet from multiple devices.
    - Example: Accessing Google Drive on a phone or laptop.
3. **Resource Pooling**
    - Shared computing resources among multiple users.
    - Example: Multiple customers using the same data center.

Characteristics of Cloud Computing)

4. **Rapid Elasticity (Scalability)**
   - Resources can scale up or down quickly.
   - Example: Adding servers during peak traffic.
5. **Measured Service (Pay-as-You-Go)**
   - Pay only for what you use.
   - Example: Paying for used storage space.
6. **High Availability**
   - Services remain available even during failures.
   - Example: Websites staying online during server failure.

Characteristics of Cloud Computing

7. **Flexibility**
    - Users can choose different configurations.
    - Example: Selecting CPU, RAM, and storage options.
8. **Cost Efficiency**
    - Reduces hardware and maintenance costs.
    - Example: Startups avoiding server purchase.

Advantages of Cloud Computing

1. **Cost Savings**
   - No need to buy expensive hardware or servers.
   - Pay only for what you use (Pay-as-you-go model).
2. **Scalability**
   - Resources can be increased or decreased instantly.
   - Handles both small and large workloads easily.
3. **Flexibility**
   - Users can choose different configurations as needed.
   - Supports multiple platforms and devices.

Advantages of Cloud Computing

4. **High Availability**
   - Services are available almost all the time.
   - Backup systems prevent downtime.
5. **Automatic Updates**
   - Software and security updates are managed by providers.
   - Reduces maintenance work for users.
6. **Accessibility**
   - Access data and applications from anywhere.
   - Works on laptops, mobiles, and tablets.

Disadvantages of Cloud Computing (1/2)

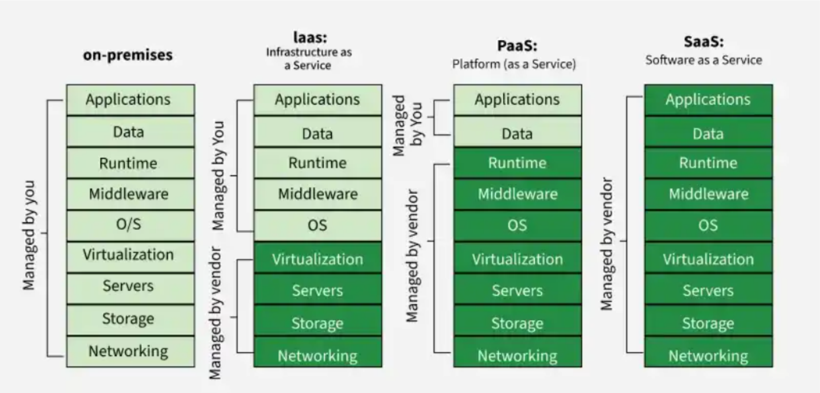
1. **Internet Dependency**
   - Cloud services require a stable internet connection.
   - No internet means no access to resources.
2. **Security and Privacy Risks**
   - Data is stored on external servers.
   - Risk of data breaches or unauthorized access.
3. **Downtime Issues**
   - Cloud services may go down due to technical failures.
   - Can affect business operations.

Disadvantages of Cloud Computing (2/2)

4. **Limited Control**
    - Users have less control over infrastructure.
    - Provider manages most system operations.
5. **Vendor Lock-in**
    - Switching providers can be difficult and costly.
    - Dependence on a single cloud provider.
6. **Long-Term Cost**
    - Continuous usage can become expensive over time.
    - Especially for large-scale workloads.

Types of Cloud Computing Services

Types Of Cloud Computing Services


on-premises
Iaas:
Infrastructure as a Service
PaaS:
Platform (as a Service)
SaaS:
Software as a Service
Managed by you
Managed by You
Managed by vendor
Managed by You
Managed by vendor
Managed by vendor
Applications
Data
Runtime
Middleware
O/S
OS
Virtualization
Servers
Storage
Networking

Infrastructure as a Service (IaaS)

Infrastructure as a Service (IaaS) is a cloud computing model that provides IT resources such as:

- Virtual machines (computing power)
- Storage systems
- Networking resources

Users access these resources over the internet without buying or managing physical hardware.

Pay only for what you use.

Benefits of IaaS

1. **Flexibility and Control**
    - Provides virtualized resources like VMs, storage, and networks.
    - Users can control operating systems and applications.
2. **Reduced Hardware Costs**
    - No need for physical infrastructure investment.
    - Reduces overall IT expenses.

Benefits of IaaS

3. **Scalability of Resources**
    - Resources can be scaled up or down based on demand.
    - Ensures optimal performance and cost efficiency.

Platform as a Service (PaaS)

Platform as a Service (PaaS) is a cloud computing model where a third-party provider offers:

- Development tools
- Software environment
- Hardware infrastructure

It allows developers to:

- Build applications
- Test applications
- Deploy applications

without managing servers or infrastructure.

Focus on code, not infrastructure.

Example of PaaS

AWS Elastic Beanstalk is an example of PaaS.

- Developers upload their application code.
- AWS automatically handles:
  - Server management
  - Load balancing
  - Scaling
- Makes deployment faster and easier.

Benefits of PaaS

1. **Simplified Development**
   - Infrastructure is hidden (abstraction layer).
   - Developers focus only on application logic.
2. **Higher Productivity**
   - Reduces infrastructure management complexity.
   - Faster development and deployment cycles.
3. **Automatic Scaling**
   - Resources scale automatically based on demand.
   - Ensures performance and efficiency.

Software as a Service (SaaS)

Software as a Service (SaaS) is a cloud computing model where software is used over the internet instead of installing it on a computer.

- Applications are hosted by a provider.
- Users access them via a web browser.
- No installation, maintenance, or storage management needed.

Use software online anytime, anywhere.

Example of SaaS

Google Docs is a common example of SaaS.

- Users can create, edit, and share documents online.
- No need to install any software.
- Data is stored in the cloud and accessible from any device.

Benefits of SaaS

1. **Collaboration and Accessibility**
    - Easy access without local installation.
    - Supports real-time collaboration over the internet.
2. **Automatic Updates**
    - Providers handle updates and maintenance.
    - Users always get the latest features and security patches.
3. **Cost Efficiency**
    - Reduces need for software licenses and IT support.
    - More affordable for individuals and businesses.